

Python Lambda Function Notes

A lambda function in Python is a small anonymous function created using the **lambda** keyword. It can take multiple arguments but contains only one expression.

Syntax

```
lambda arguments : expression
```

Examples

1. Square Function

```
square = lambda x: x * x  
print(square(5))  
Output: 25
```

2. Addition Function

```
add = lambda a, b: a + b  
print(add(2, 3))  
Output: 5
```

3. Using map()

```
numbers = [1,2,3,4]  
result = list(map(lambda x: x*2, numbers))  
Output: [2,4,6,8]
```

4. Using filter()

```
numbers = [1,2,3,4,5,6]  
even = list(filter(lambda x: x%2==0, numbers))  
Output: [2,4,6]
```

Important Points

Lambda functions are short anonymous functions.
They are mainly used for temporary operations.
They are commonly used with map(), filter(), and sorted().
They can have multiple arguments but only one expression.